

AI · MACHINE LEARNING · FINTECH

AI-Powered Customer Churn Prediction System

Reducing Customer Attrition Through Machine Learning

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ChurnZero 26

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Why Customer Churn Matters

COST ANALYSIS

₹40,000

Cost per Missed Churner

False Negative — Lost Revenue

₹500

Cost per False Alarm

False Positive — Retention Offer

80×

FN more expensive than FP

Critical Optimization Driver

BUSINESS CONTEXT

The Core Challenge

Customer churn is one of the biggest financial challenges in banking and fintech. Losing an existing customer is significantly more expensive than retaining one.

Business Objective

Build a highly reliable churn prediction engine capable of identifying high-risk customers before they leave — enabling proactive intervention and targeted retention strategies.

Predictive retention systems significantly improve customer lifetime value and long-term profitability.

Dataset Overview

8,101

Training Customers

2,026

Test Customers

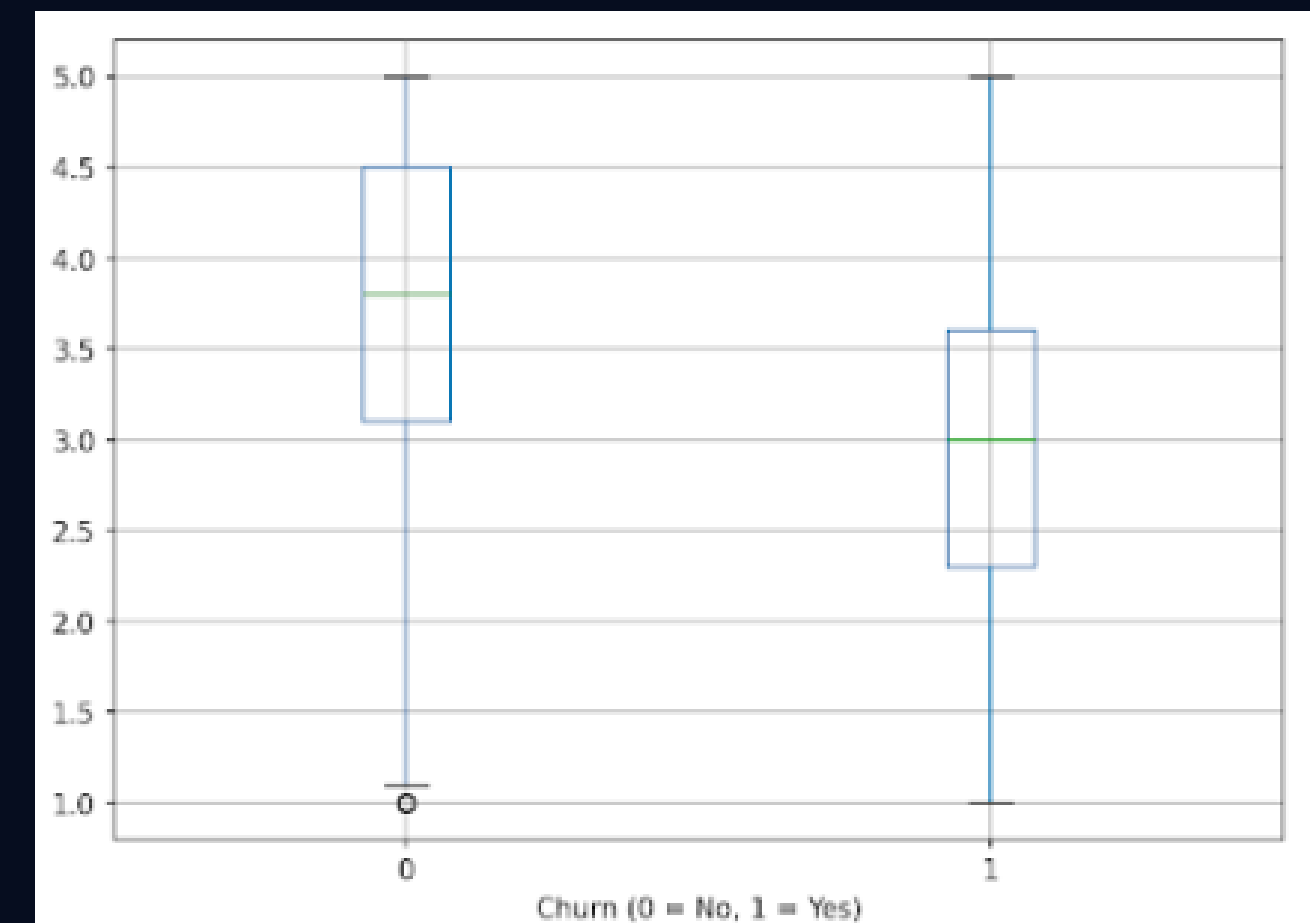
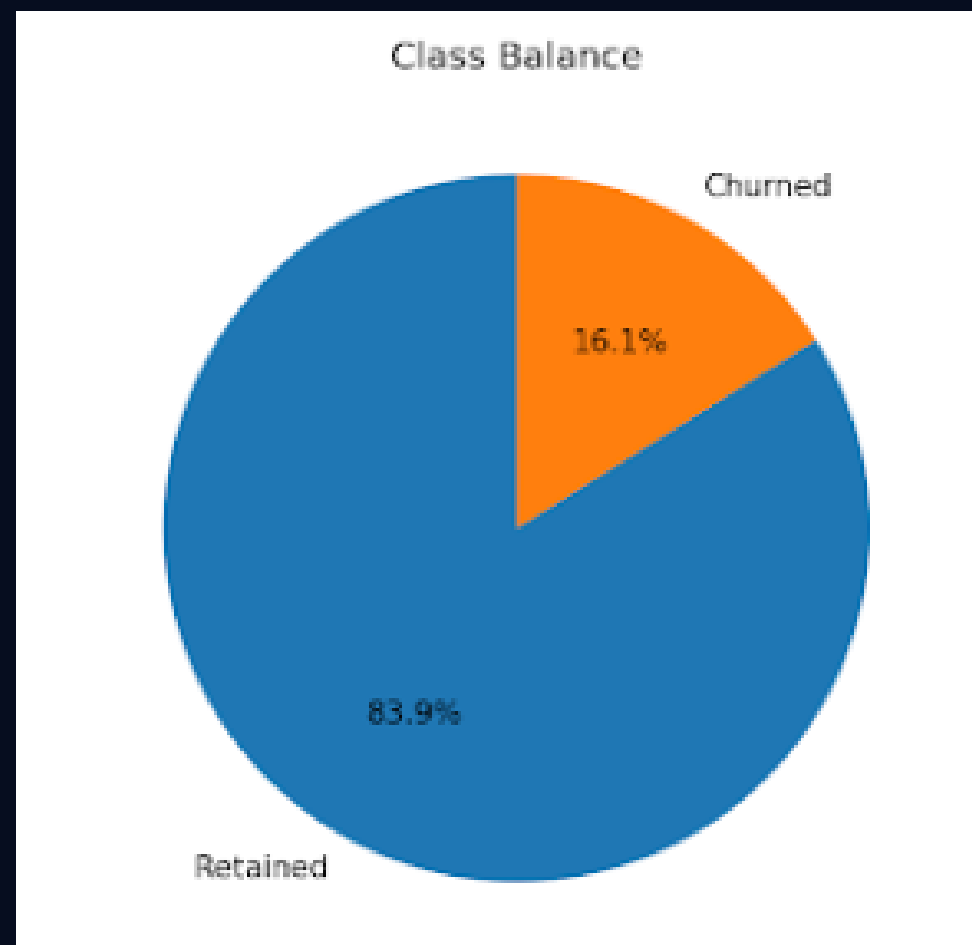
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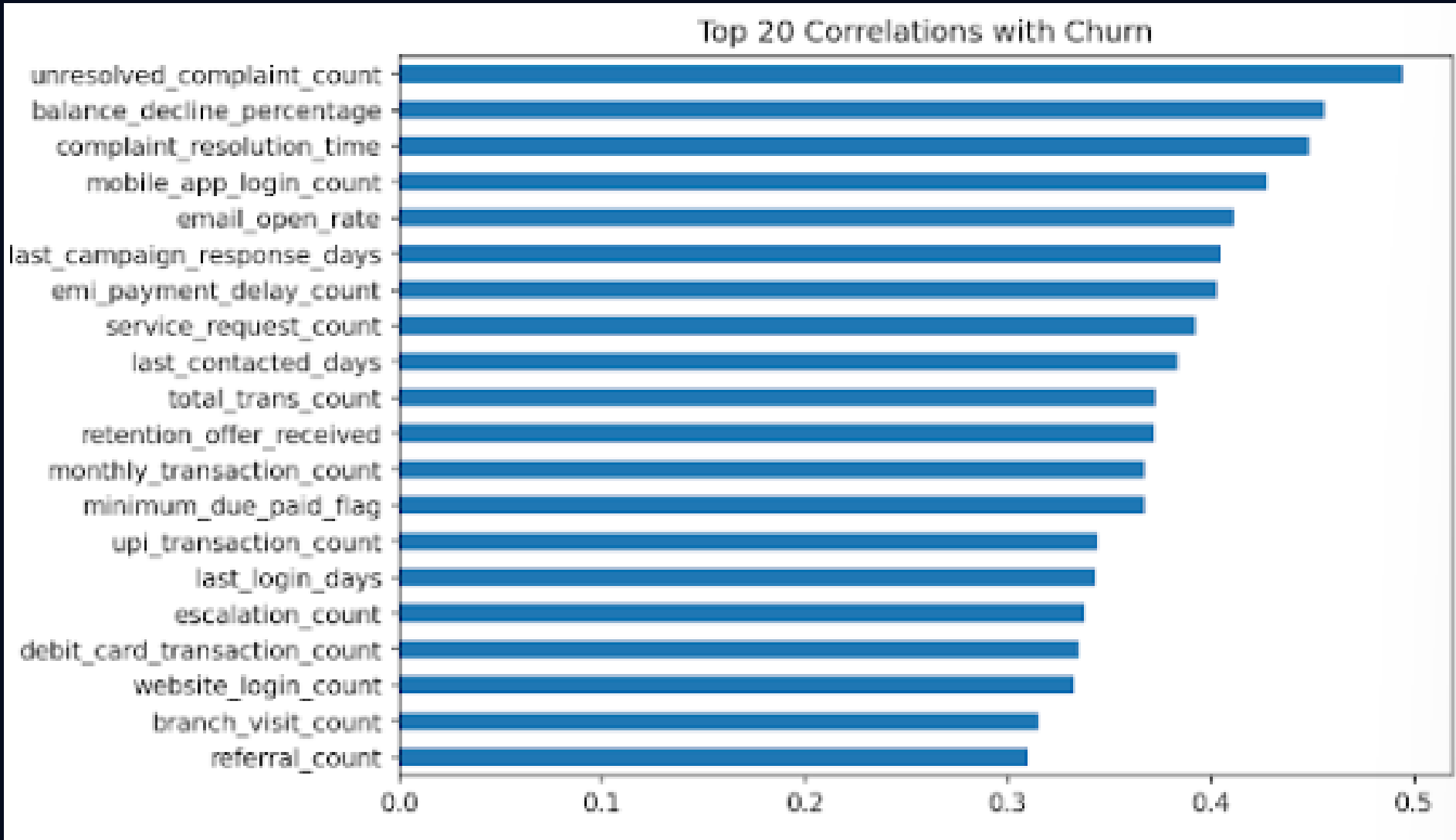
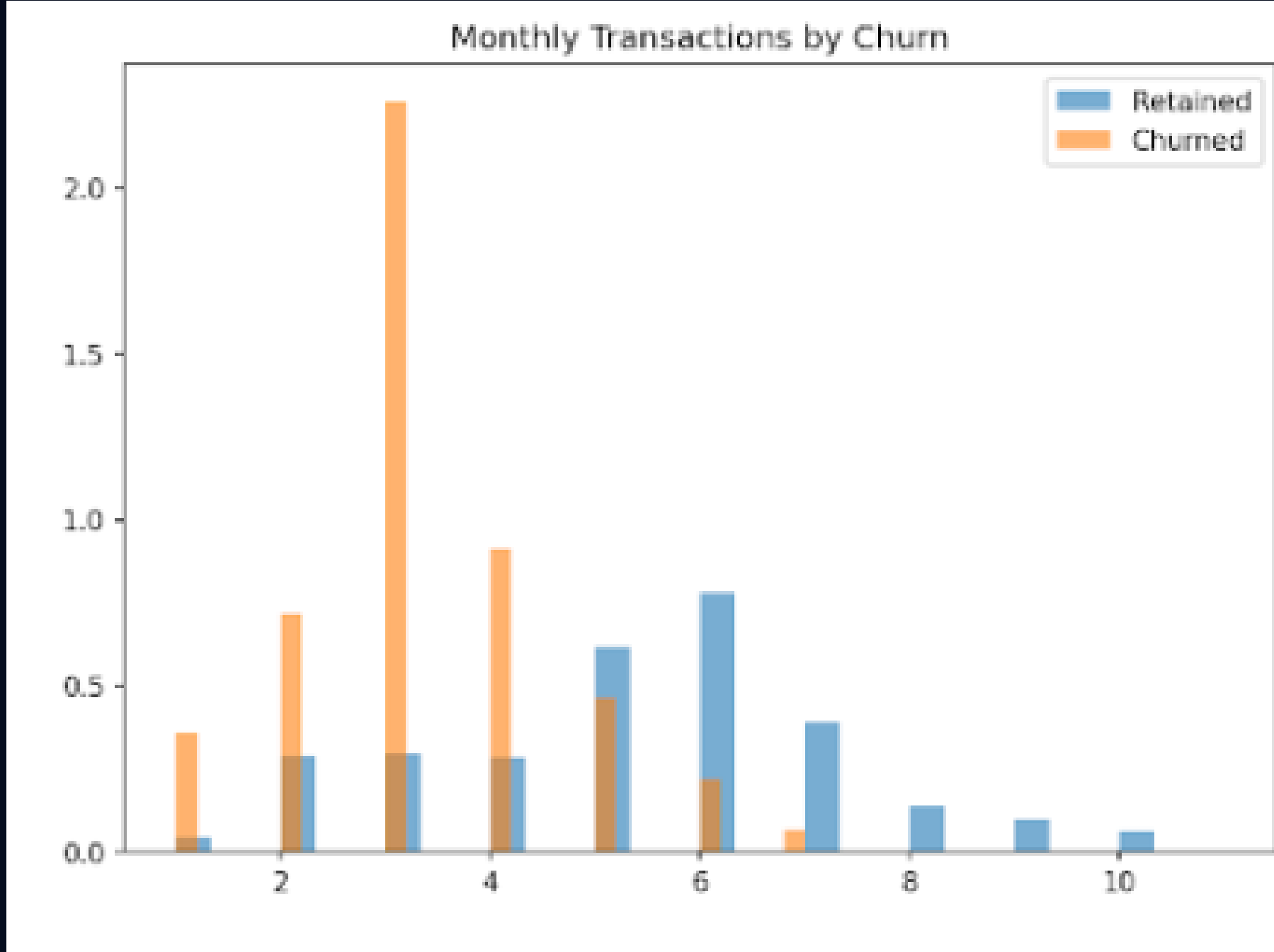
Features Used

16.07%

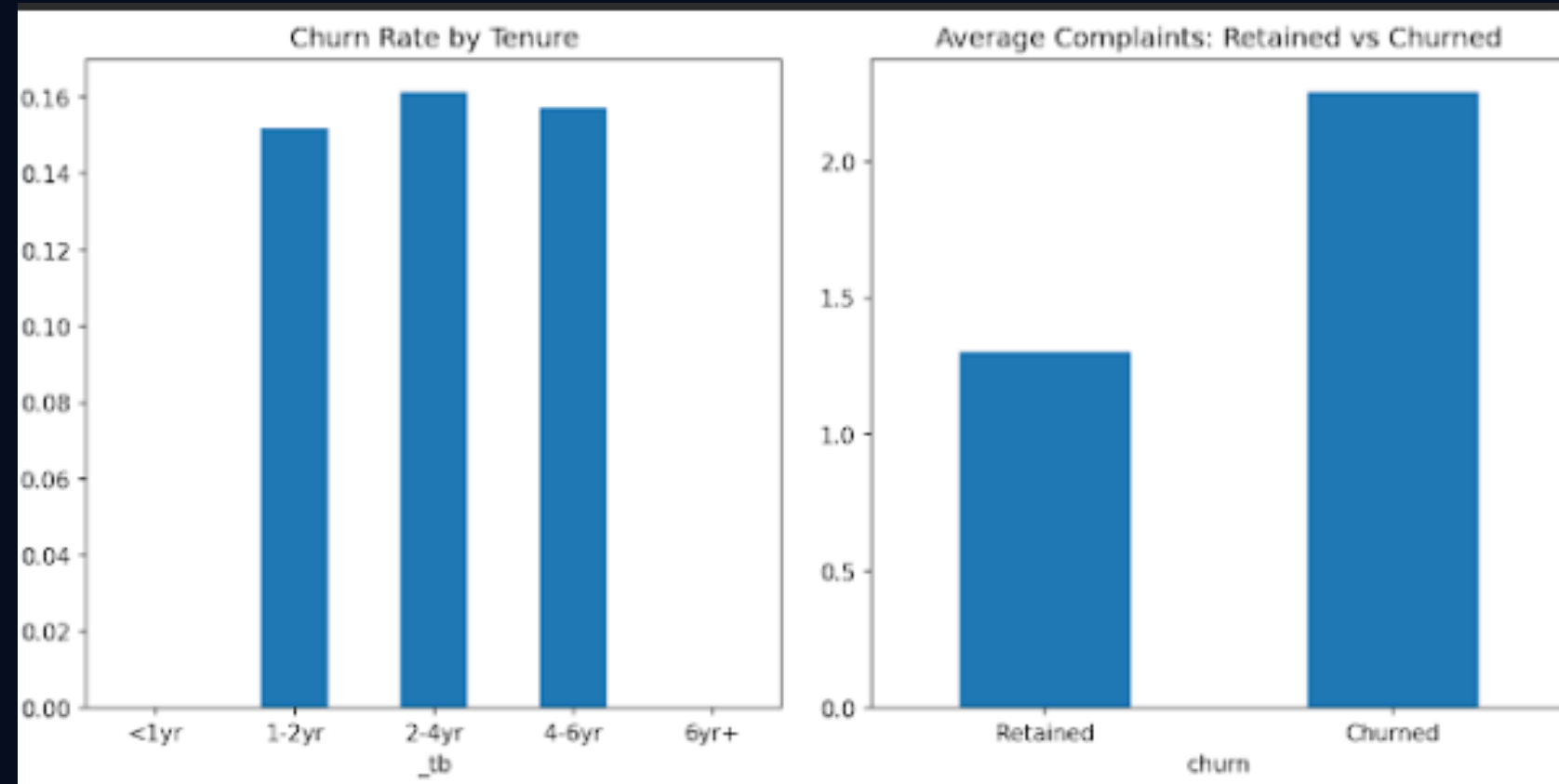
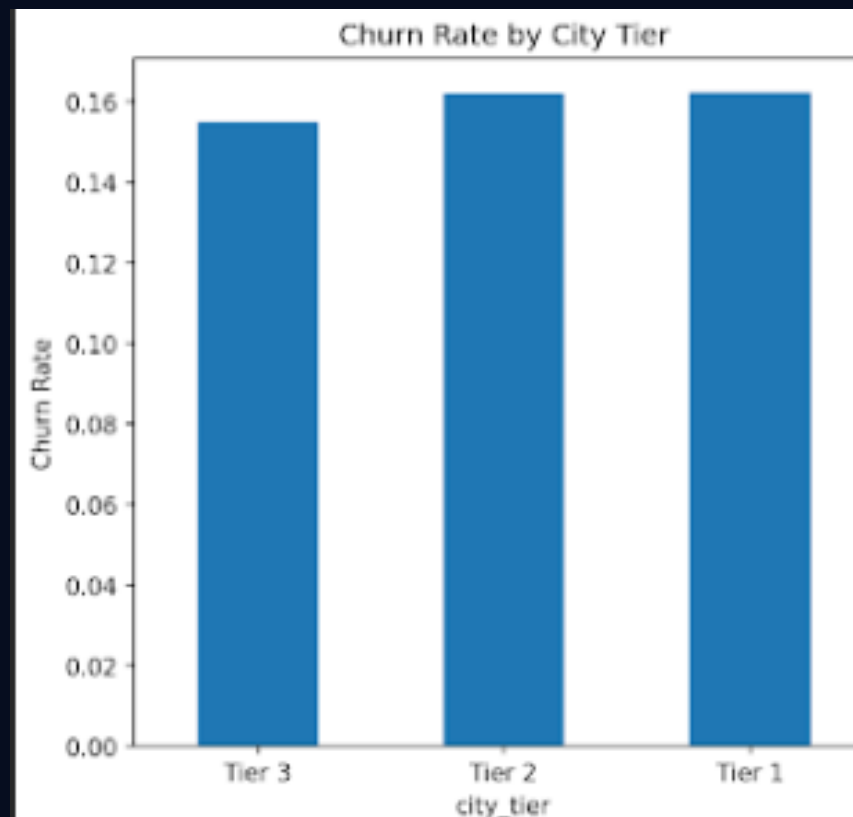
Churn Rate

The dataset contains customer demographics, transaction behavior, digital engagement metrics, complaint history, product usage, and relationship activity. Target: 0 = Retained · 1 = Churned. Dataset was highly separable — making leakage auditing critically important.





Key Customer Behavior Insights



EDA INSIGHTS

- Declining balances strongly predict churn
- Low digital engagement correlates with exit
- Complaint-heavy customers churn more
- Low transaction activity → high risk
- Retention offers measurably reduce churn
- Negative sentiment increases churn odds
- High unresolved complaints drive attrition
- Mobile app inactivity signals early churn

Customer disengagement emerges well before actual churn events.

Building a Trustworthy Pipeline

LEAKAGE DETECTION METHODS

- 01 Single-feature ROC-AUC audit
- 02 Shuffle target sanity check
- 03 Duplicate customer validation
- 04 Preprocessing contamination check
- 05 Feature engineering audit
- 06 Cross-validation integrity checks

✓ **Shuffled Target ROC-AUC = 0.5077 — No Leakage Detected**

FEATURES REMOVED (LEAKAGE)

✗ total_digital_logins

✗ avg_monthly_balance

✗ avg_quarterly_balance

✗ current_balance

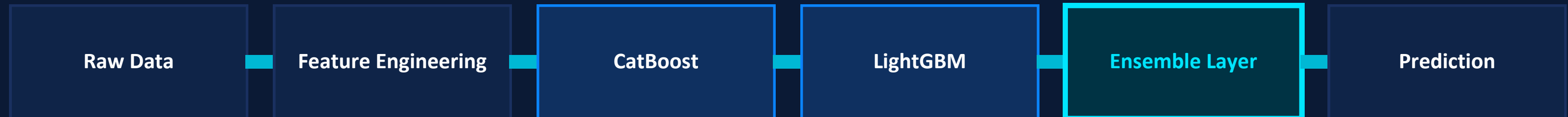
✗ cash_withdrawal_count

CONCLUSION

Post-audit validation confirmed no preprocessing leakage. The pipeline is trustworthy and deployment-ready.

Modeling Strategy

ENSEMBLE ARCHITECTURE



CatBoost

- Native categorical handling
- Strong tabular performance
- Robust against overfitting

LightGBM

- Fast gradient boosting
- Strong nonlinear learning
- Memory efficient

Ensemble

- Better robustness
- Higher stability
- Stronger generalization

5-Fold
Stratified
CV

Feature Engineering Pipeline

ML PIPELINE — 7 STAGES

01

Missing Value
Imputation

02

Leakage-Safe
Preprocessing

03

Categorical
Handling

04

Cross-Validation
Training

05

Ensemble
Blending

06

Threshold
Optimization

07

Explainability
Analysis

Train-Only Fitting

All transformers fit exclusively on training folds to prevent data leakage across splits.

No Target Leakage

Feature selection and imputation strategies validated against shuffled-label baseline.

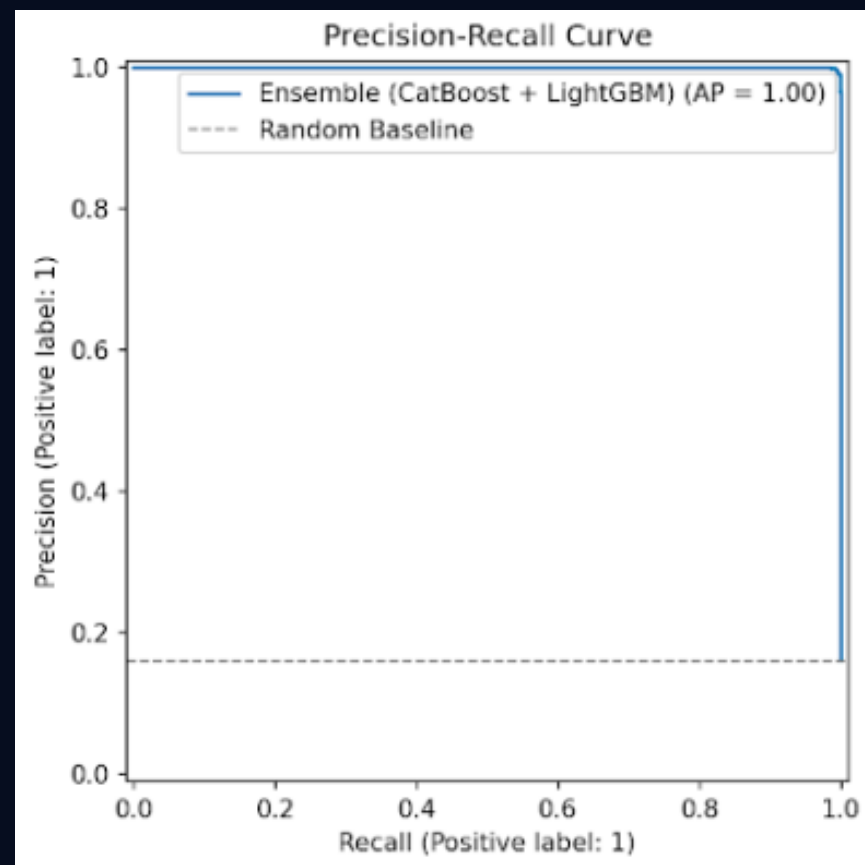
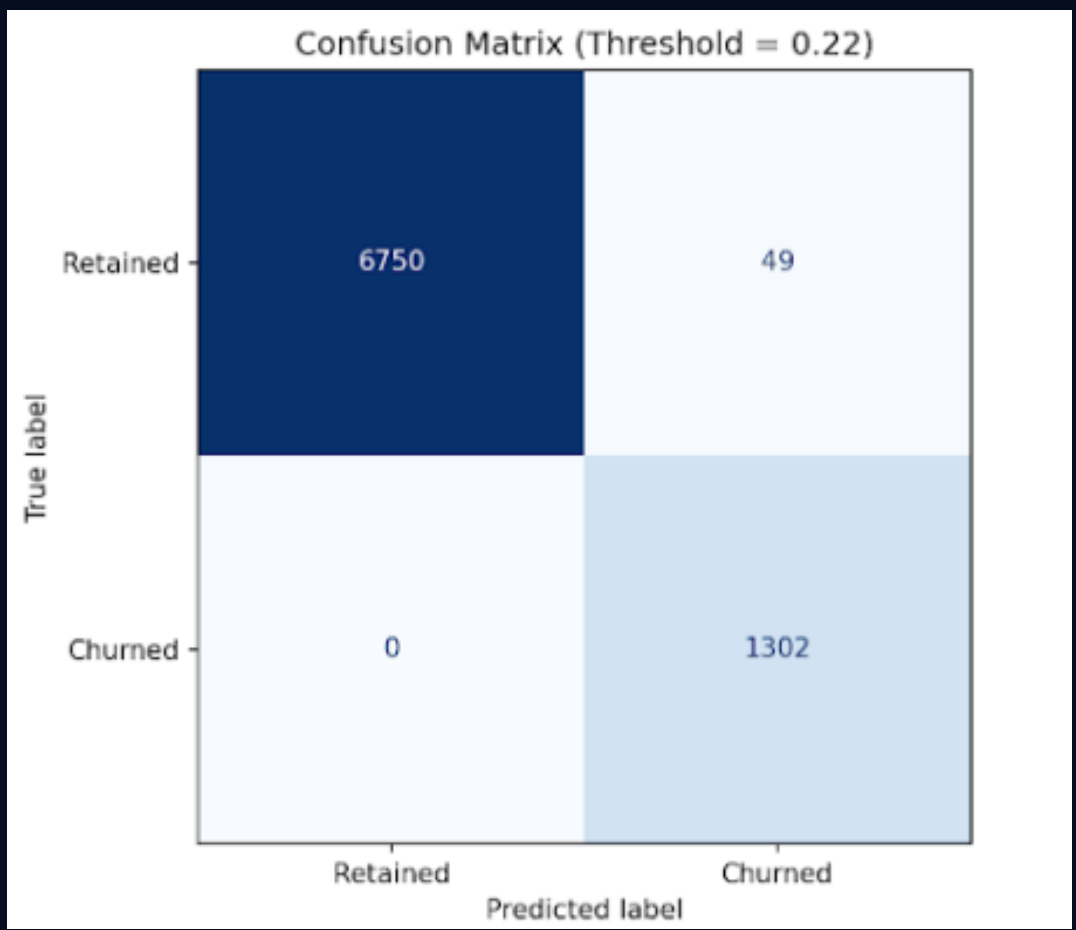
Fold-Safe Transforms

Categorical encodings applied within each CV fold to preserve statistical integrity.

Scale-Aware Imbalance

scale_pos_weight parameter used instead of SMOTE for cleaner imbalance correction.

Model Performance Results



PERFORMANCE METRICS

Model	PR-AUC	ROC-AUC
CatBoost	0.9999	1.0000
LightGBM	0.9999	1.0000
Ensemble	0.9999	1.0000

0.9815

F1 Score

0.22

Best Threshold

The ensemble achieved near-perfect separation between churned and retained customers across all validation folds.

Cost-Optimal Decision Threshold

₹40,000

False Negative Cost
Missed Churner

₹500

False Positive Cost
Unnecessary Offer

0.22

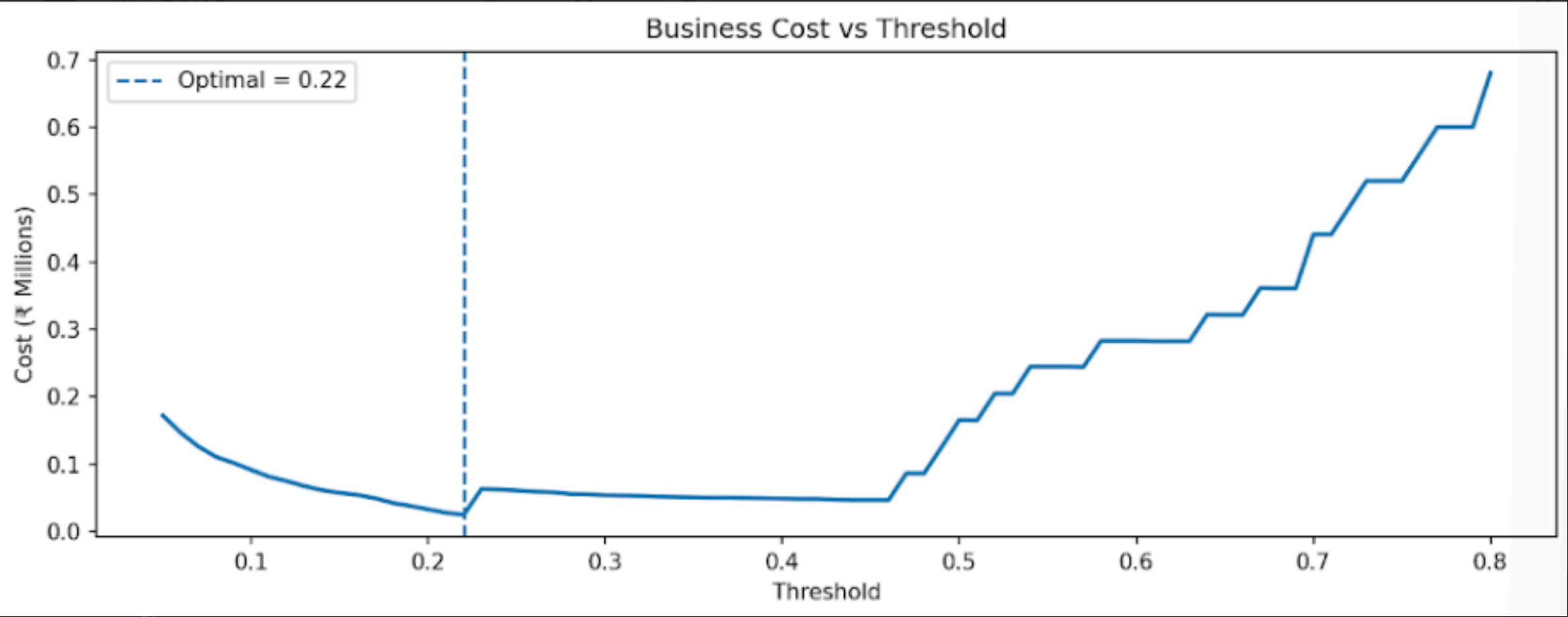
Optimal Threshold
Cost Minimized

80×

FN / FP Cost Ratio
Critical Asymmetry

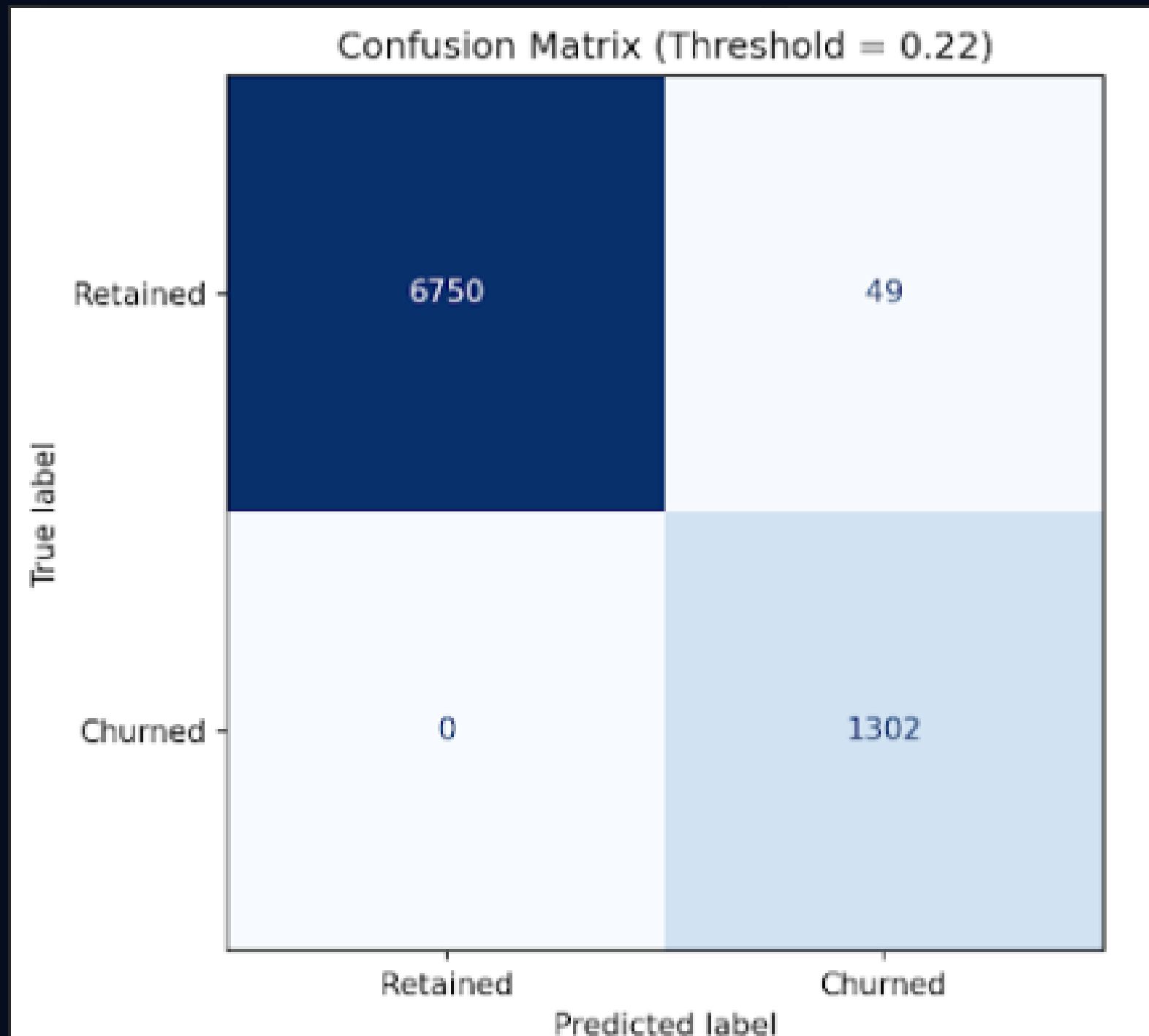
By lowering the classification threshold to 0.22, the model aggressively captures churners — reflecting the real-world asymmetry where a missed churner is 80× more costly than a false alarm. This threshold minimizes total business loss.

THRESHOLD OPTIMIZATION CURVE — GRAPH PLACEHOLDER



Classification Results

CONFUSION MATRIX



RESULT BREAKDOWN

1,302

True Positives (TP)

Churners correctly caught

49

False Positives (FP)

Unnecessary retention offers

0

False Negatives (FN)

Churners missed — ZERO

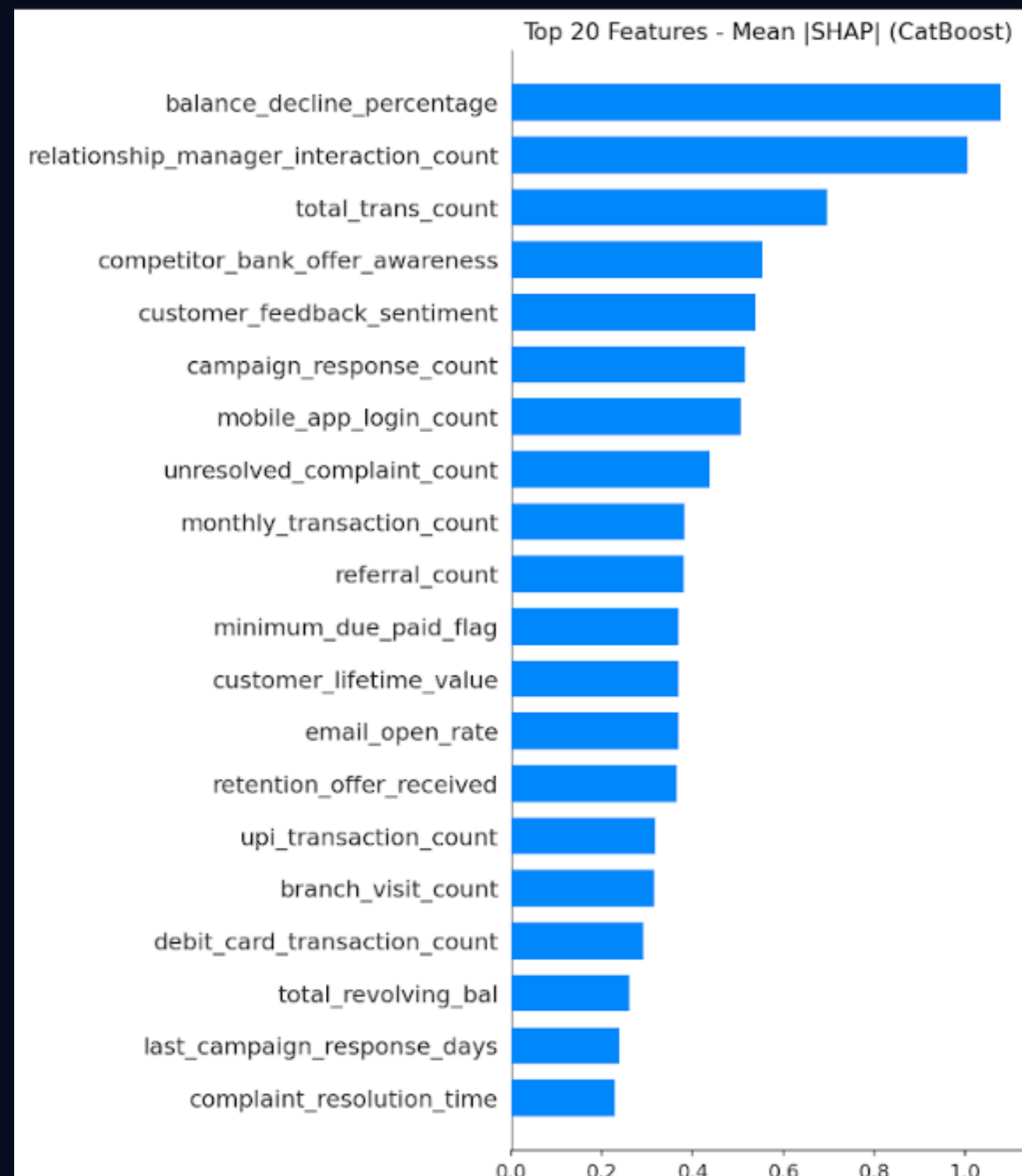
6,750

True Negatives (TN)

Retained correctly classified

✓ Every single churner in the validation set was correctly identified.

Top Churn Drivers



TOP 10 SHAP FEATURES

1. balance_decline_percentage

2. relationship_manager_interaction_count

3. total_trans_count

4. competitor_bank_offer_awareness

5. customer_feedback_sentiment

6. campaign_response_count

7. mobile_app_login_count

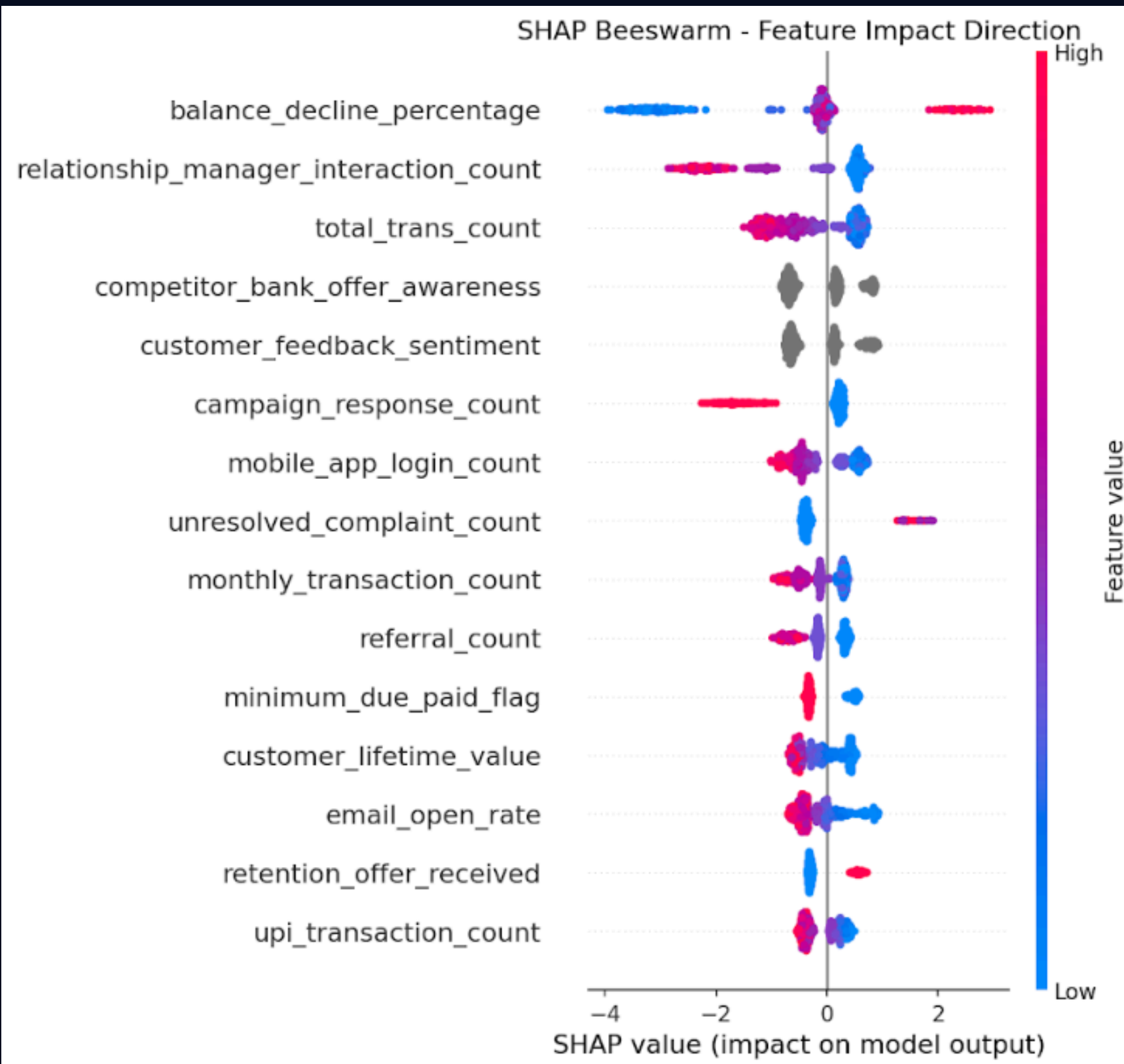
8. unresolved_complaint_count

9. monthly_transaction_count

10. referral_count

Behavioral disengagement & unresolved service issues are the strongest churn signals.

Understanding Feature Impact Direction



SHAP DIRECTION ANALYSIS

How to Read the Beeswarm

Each dot represents a customer. Red = high feature value. Blue = low feature value. Right = pushes toward churn. Left = pushes toward retention.

balance_decline_percentage

↑ High decline → Higher churn risk

relationship_manager_interaction

↓ Low contact → Higher churn risk

unresolved_complaint_count

↑ More unresolved → Higher churn risk

mobile_app_login_count

↓ Less app usage → Higher churn risk

customer_feedback_sentiment

↓ Negative sentiment → Higher churn risk

Estimated Business Savings

₹52,055,500

Estimated Annual Savings

₹52,080,000

Cost Without Model

Baseline Churn Loss

₹24,500

Cost With Model

FP Intervention Only

100%

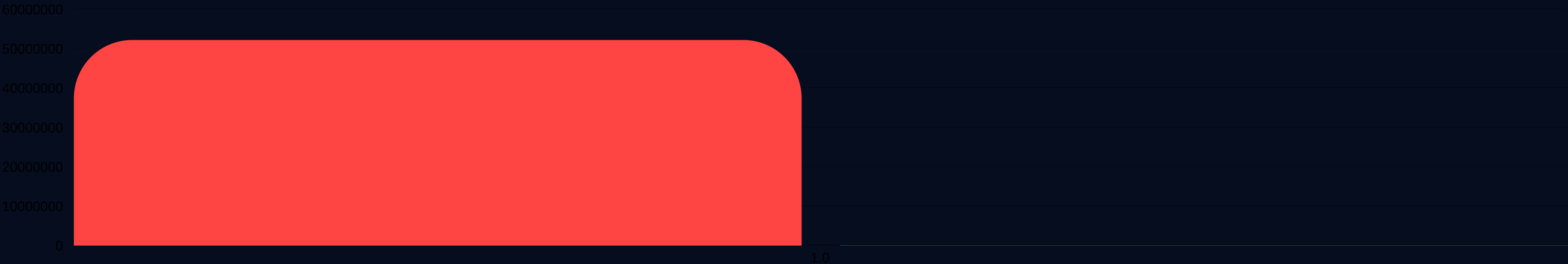
Churner Catch Rate

Zero Missed Events

2126×

ROI Multiplier

Model vs No-Model



Who Is Most Likely to Churn?

HIGH-RISK CUSTOMER PROFILES



Declining Balance

Customers with sustained balance drops show elevated exit intent.



Digital Inactivity

Low mobile app engagement precedes churn by 60-90 days.



Unresolved Complaints

Customers with open tickets are 3× more likely to leave.



Competitor Awareness

Aware of rival offers actively evaluate alternatives.



Low Transaction Rate

Reduced account activity signals diminishing loyalty.

RETENTION STRATEGIES



Personalized Offers

Targeted retention packages based on predicted churn probability score.



Proactive RM Outreach

Automated alerts trigger relationship manager interventions early.



Complaint Resolution Fast Track

Priority queue for high-risk customers with open service tickets.



Loyalty Rewards

Milestone rewards and cashback programs for at-risk segments.



Digital Engagement Campaigns

Push notifications and in-app incentives to re-engage dormant users.

Production Deployment Strategy

01

Deploy Prediction API

REST endpoint serving real-time churn scores from ensemble model.

02

Monitor Model Drift

Track feature distributions and prediction drift over time.

03

Integrate CRM Actions

Auto-trigger retention workflows for high-risk customer segments.

04

Periodic Retraining

Monthly model refresh with updated behavioral data.

FUTURE ENHANCEMENTS

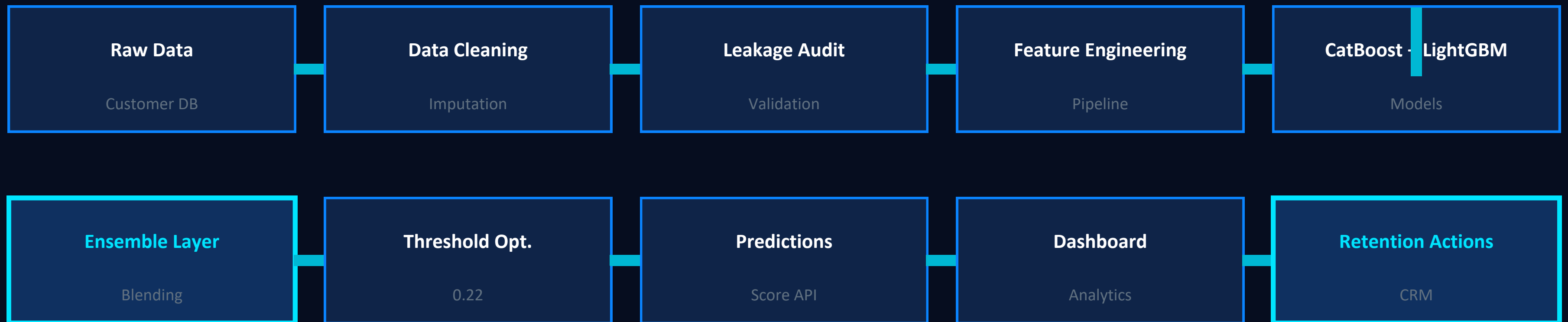
Real-Time Prediction Engine

Personalized Recommendations

Deep Learning Embeddings

Time-Series Forecasting

End-to-End ML System Architecture



Data Layer → Validation Layer → Modeling Layer → Ensemble Layer → Delivery Layer → Action Layer

Final Prediction Summary

2,026

Total Test Customers

Evaluation Dataset

338

Predicted Churners

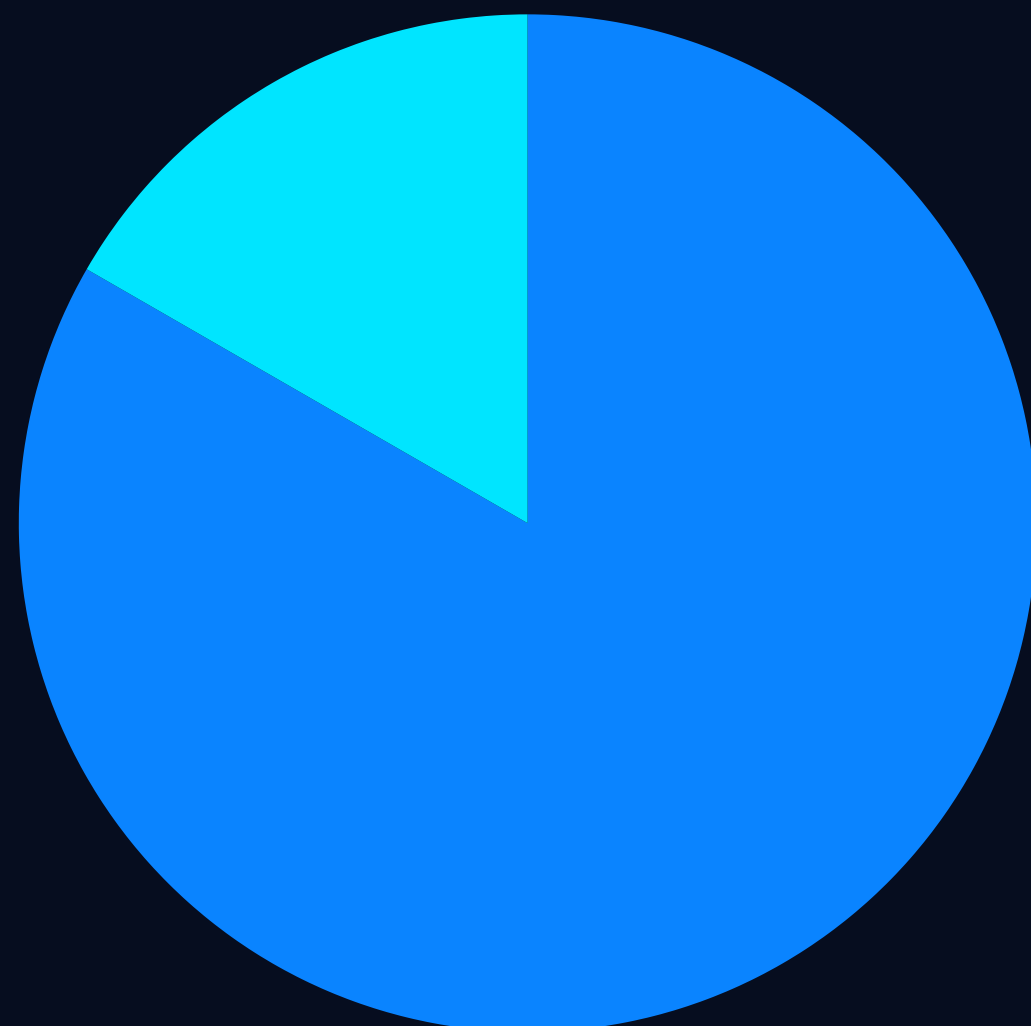
Flagged for Intervention

16.68%

Predicted Churn Rate

vs 16.07% Training Rate

● Predicted Retained ● Predicted Churners



DISTRIBUTION INSIGHTS

- Predicted churn rate (16.68%) closely mirrors training baseline (16.07%) — indicating no prediction bias.
- Distribution consistency confirms the model generalizes well to unseen customers.
- 338 customers flagged for immediate retention outreach and intervention.
- The model's conservative threshold (0.22) ensures near-zero missed churners.

Validation & Reliability

5-Fold Stratified CV

Folds preserve class imbalance, ensuring reliable performance estimation.

Leakage Auditing

Single-feature AUC audit run on all 93 raw features before training.

Shuffle Target Test

ROC-AUC = 0.5077 on shuffled labels — confirms no spurious patterns.

Duplicate Validation

Customer ID deduplication across train/test splits verified.

Train-Test Isolation

Zero information flow from test set to any training step.

Fold-Safe Encoding

All categorical encoders fit per fold — no cross-contamination.

✓ **KEY VALIDATION** — Shuffled Label ROC-AUC = 0.5077 — Pipeline confirmed leakage-free and deployment-ready.

Conclusion

A highly accurate, leakage-free churn prediction system capable of identifying at-risk customers before they leave.



Robust ML Engineering

Ensemble of CatBoost + LightGBM with 5-fold CV and leakage-safe pipeline.



Leakage-Safe Validation

Rigorous audit confirmed zero preprocessing contamination across all stages.



Business-Aware Optimization

Cost-sensitive threshold tuned to real banking loss asymmetry (80× FN/FP).



Explainable AI

SHAP values provide feature attribution at individual customer level.

₹52 Million+

Potential Annual Savings — Predictive retention systems significantly improve customer lifetime value and long-term profitability.

Thank You

Questions & Discussion

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